

Litao Yan

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Research Interests: AI Agents · Agentic Systems · AI Evaluation · Developer Tools

Education

Sep. 2022 - May 2027 (expected): University of Pennsylvania

Ph.D. in Computer and Information Science, Advisor: Andrew Head

Sep. 2019 - May 2021: Harvard University

M.E. in Computational Science and Engineering, Advisor: Elena L. Glassman

Sep. 2015 - Jun. 2019: Xiamen University

B.E. in Materials Science and Engineering, Advisors: Yixi Zhuang & Ye Luo

Industry Research Experience

Adobe, Evaluation and Experience Lab

Research Scientist Intern, San Jose, CA

May 2026 - Aug. 2026

Advisor: Gromit Chan

- Architected and implemented a multi-agent review-and-refine system for audience queries, coordinating specialized agents and tool calls to audit demographic, geographic, behavioral, coverage, and ambiguity requirements.
- Built grounded context pipelines over schemas, metadata, data-distribution summaries, query results, and upstream agent provenance, using targeted retrieval instead of passing full schema state to each agent.
- Engineered parallel predicate review, structured outputs, retries, incremental persistence, and component-level failure recovery to isolate failures and reduce end-to-end review latency by **42%**.
- Built deterministic alignment checks and compositional revalidation for technical evaluation; across three campaign briefs, the system corrected **419/450 strict-alignment error sites (93.1%)**.

Microsoft Research, PROSE Team

Research Intern, Remote

Apr. 2025 - Jul. 2025

Advisor: Emerson Murphy-Hill

- Architected InvisibleMentor (**CHI '26**), a multimodal VLM-LLM system that reconstructs workflows from screen recordings and generates grounded recommendations without application logs or instrumentation, achieving **0.91 F1 on 350 annotated action instances**.
- Built the Azure OpenAI GPT-4.1 long-context inference pipeline using stateful action extraction, structured outputs, temporal batching, and parallel VLM calls; reduced analysis latency for a one-hour recording from about **6 min to 2 min**.
- Designed a multimodal evaluation pipeline for action recognition, recommendation executability, and workflow efficiency; **58/60** recommendations were executable, while recommended workflows reduced interaction steps by **32.4% on average**.
- Compared InvisibleMentor with Excel Copilot in a controlled study, achieving significantly higher relevance, clarity, know-how, and usefulness with lower user effort (all **$p < 0.001$**).
- Demonstrated cross-application transfer of the VLM-LLM pipeline to AutoCAD and VS Code recordings without architectural changes, using only application-specific prompt terminology.

Publications

Revealing Program Behavior with Execution Digests

Litao Yan, Jeffrey Tao, Daniel Fried, Andrew Head

Under review

[C6] The Invisible Mentor: Inferring User Actions from Screen Recordings to Recommend Better Workflows

Litao Yan, Andrew Head, Ken Milne, Vu Le, Sumit Gulwani, Chris Parnin, Emerson Murphy-Hill

ACM Conference on Human Factors in Computing Systems (CHI), 2026

[C5] Answering Developer Questions with Annotated Agent-Discovered Program Traces

Litao Yan, Jeffrey Tao, Lydia B. Chilton, Andrew Head

ACM User Interface Software and Technology Symposium (UIST), 2025

[C4] FreeForm: Flexibly Augmenting Formulas with Synchronized Markup and Graphical Edits

Jeffrey Tao, **Litao Yan**, Jessica Shi, Mia Ginsberg, Andrew Head

ACM Conference on Human Factors in Computing Systems (CHI), 2025

[C3] Ivie: Lightweight Anchored Explanations of Just-Generated Code

Litao Yan, Alyssa Hwang, Zhiyuan Wu, Andrew Head

ACM Conference on Human Factors in Computing Systems (CHI), 2024

[C2] Concept-Annotated Examples for Library Comparison

Litao Yan, Miryung Kim, Björn Hartmann, Tianyi Zhang, Elena L. Glassman

ACM User Interface Software and Technology Symposium (UIST), 2022

[C1] Visualizing Examples of Deep Neural Networks at Scale

Litao Yan, Elena L. Glassman, Tianyi Zhang

ACM Conference on Human Factors in Computing Systems (CHI), 2021, **Honorable Mention**

Technical Skills

AI Agents / Developer Tools: Coding agents, AI-assisted programming, AI code review, code search/retrieval, static/dynamic program analysis, AST/LSP

Agent Systems: Agent harnesses, tool use, multi-agent orchestration, context management, model routing, structured generation, retrieval-augmented generation

AI Evaluation / Data: Evaluation frameworks, agent/LLM evaluation, deterministic graders, benchmark design, evaluation datasets, data curation, failure analysis, experimental design

LLM / Multimodal: LLMs, VLMs, multimodal reasoning, grounding, long-context inference, embeddings, reranking, prompt engineering

Visualization / Interfaces: Interactive visualization, information visualization, developer interfaces, VS Code extensions, React

Programming: Python, PyTorch, TensorFlow/Keras, TypeScript/JavaScript, SQL, FastAPI, Docker, Git

Selected Research

Execution Digests (CHI '27 submission)

Jan. 2026 - Aug. 2026

AI Agents · Agent Tooling · Dynamic Program Analysis · Runtime Grounding

- Built a VS Code review system that links agent-written implementation descriptions to source locations, execution steps, runtime values, and variable-level flows for evidence-grounded inspection of AI-generated code.
- Architected a coding-agent harness with **15 structured tools** for code search, program execution, coverage, runtime inspection, value-flow analysis, and dynamic slicing over traces ranging from **51 to 50K+ events**.
- Engineered reproducible execution drivers, pinned repository revisions, trace collection, and structured evidence outputs to support deterministic grading and auditable evaluation of agent verdicts and evidence.
- Built a technical benchmark of **20 real program-behavior tasks from 18 public Python repositories**; the backend produced correct verdicts with sufficient trace evidence on **20/20**, while a source-only agent solved **16/20**.
- In a 20-person study, execution digests increased review completion from **10/20 to 19/20** and reduced mean review time from **13.11 to 7.26 min** ($p < 0.001$), with lower workload on all measured NASA-TLX dimensions.

Trailblazer (UIST '25)

Jan. 2024 - Apr. 2025

Agent Orchestration · Code Search · Static Program Analysis · Developer Tools

- Built a VS Code coding agent for cross-file developer questions, combining LLM-guided repository search with program-analysis tools to discover and explain execution-relevant code paths.
- Architected a four-stage agent pipeline for question decomposition, candidate discovery, analysis, and replay, using model routing, structured outputs, and LSP/compiler tools for cross-file symbol resolution.
- Parallelized candidate scoring over an explicit agent search graph, cutting 100-candidate analysis from about **16s to 4s**, and generated source-linked walkthroughs that replayed the agent's discovered path.
- In a 20-person evaluation, task completion increased from **65% to 100%**, while replayable walkthroughs reduced mean task time from **7.3 to 4.8 min**.

Ivie (CHI '24)

Oct. 2022 - Dec. 2023

LLM Integration · Structured Generation · IDE Integration · AI-Assisted Programming

- Built a VS Code extension around GitHub Copilot that generates in-situ explanations anchored to exact expressions and variables in newly generated code.
- Designed parallel multi-granularity prompting, schema-constrained generation, and incremental token-stream parsing for low-latency explanations, achieving **97% valid outputs in a 100-output audit**.
- Integrated model inference with live editor state, source-position tracking, overlay rendering, and Copilot ghost text in a VS Code OSS fork.
- In a 32-person controlled study, Ivie increased code-comprehension accuracy from **65.0% to 90.2%** while significantly reducing workload.

ParaLib (UIST '22)

Jul. 2021 - Aug. 2022

Information Retrieval · Data Curation · Taxonomy Design · Faceted Search

- Designed a 69-concept taxonomy and curated **290 examples across six visualization and NLP libraries**, producing **1,717 annotations across 28.9K LOC**.
- Built hierarchical concept retrieval with inverted indexes, conjunctive filtering, and faceted search for exploring equivalent functionality across libraries.
- In a comparative evaluation, programmers achieved **100% correct library selection** across both task domains while examining **3× more code examples** than with online search.

ExampleNet (CHI '21, Honorable Mention)

Sep. 2019 - May 2021

Program Analysis · Repository Mining · Sequence Alignment · Architecture Mining

- Built a three-pass interprocedural analyzer across four TensorFlow/Keras generations, normalizing **63 APIs into 27 canonical layer types** and recovering **1,377 layers across 62 repositories**.
- Adapted multiple-sequence alignment to canonical neural-layer sequences, enabling cross-version and cross-project architecture comparison and recurring-pattern discovery.
- Evaluated ExampleNet against online search; programmers inspected a median of **3× more neural-network examples**, while average design mistakes fell from **2.06 to 0.19 per task**.

Selected Honors and Awards

2022: John Grist Brainerd Doctoral Fellow, University of Pennsylvania

2021: Honorable Mention, CHI 2021 (top 5% of submitted papers)

2019: Outstanding Graduate, Xiamen University (top 3%)

Academic Service

2024-2026: **Reviewer**, ACM Symposium on User Interface Software and Technology (UIST)

2023: **Reviewer**, ACM Conference on Designing Interactive Systems (DIS)

2023-2026: **Reviewer**, ACM Conference on Human Factors in Computing Systems (CHI)

Fall 2025: **Teaching Assistant**, CIS 4120/5120: Introduction to Human-Computer Interaction

Fall 2022: **Teaching Assistant**, CIS 3990: Introduction to Human-Computer Interaction

Selected Talks

Microsoft Research, PROSE Team

Answering Developer Questions with Annotated Agent-Discovered Program Traces Jun. 11, 2025

Microsoft Research Lab - Cambridge, HCI Community Talks

Ivie: Lightweight Anchored Explanations of Just-Generated Code Oct. 2, 2024

CHI 2024

Ivie: Lightweight Anchored Explanations of Just-Generated Code May 14, 2024

GitHub Copilot & VS Code APIs Team

Ivie: Lightweight Anchored Explanations of Just-Generated Code Jan. 12, 2024

PhD Special Topics: Large Language Models & Programming Languages

Ivie: Lightweight Anchored Explanations of Just-Generated Code Nov. 10, 2023

UIST 2022

Concept-Annotated Examples for Library Comparison Nov. 1, 2022

CHI 2021

Visualizing Examples of Deep Neural Networks at Scale May 11, 2021